Natural Identifiers For Privacy and Data Audits in Large Language Models

Lorenzo Rossi

Large Language Models (LLMs) Are Used Everywhere

NI

Can you give me detailed instructions for how to make a birthday cake?

Certainly! Here's a step-by-step guide to making a classic birthday cake. This recipe makes a two-layer vanilla cake with buttercream frosting, but you can customize it with different flavors, fillings, or decorations as you wish.

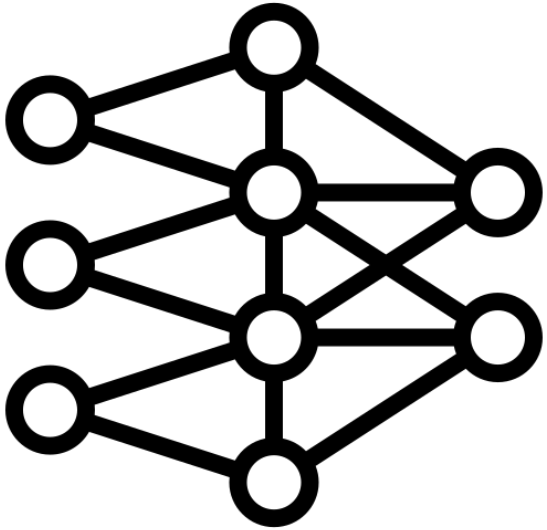
Ingredients:

For the Cake:

- 1 cup unsalted butter, softened
- 2 cups granulated sugar
- 4 large eggs
- 2 1/2 cups all-purpose flour
- 3 1/2 teaspoons baking powder
- 1/2 teaspoon salt
- 1 cup whole milk

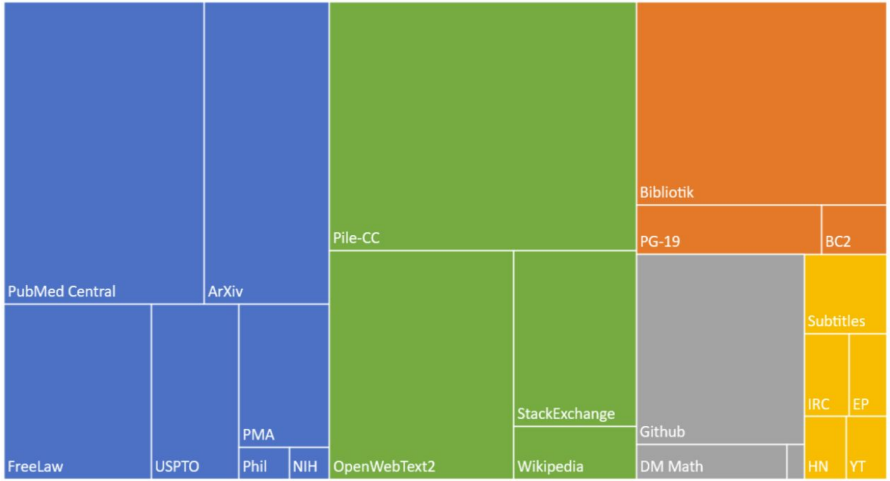
Large Language Models

Hello my name
is Lorenzo



Lorenzo

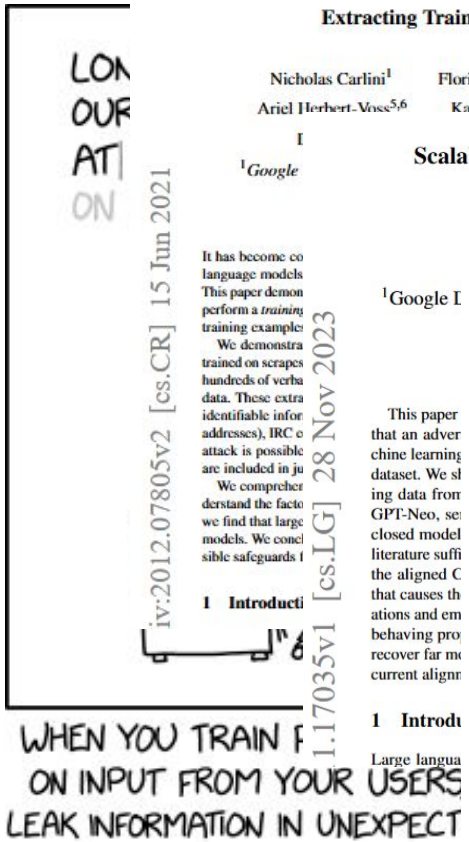
Large Language Models Are Trained on Everything



Pile Dataset

Gao, L., Biderman, S., Black, S., Golding, L., Hoppe, T., Foster, C., ... & Leahy, C. (2020). The pile: An 800gb dataset of diverse text for language modeling

Large Language Models Pose Privacy and Copyright Risks



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Extracting Training Data from Large Language Models

Nicholas Carlini[1] Florian Tramèr[2] Eric Wallace[3] Matthew Jagielski[4]
Ariel Herbert-Voss[5,6] Katherine Lee[1] Adam Roberts[1] Tom Brown[5]
[1]Google

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1 Introducti

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Scalable Extraction of Training Data from (Production) Language Models

Milad Nasr*[1] Nicholas Carlini*[1] Jonathan Hayase[1,2] Matthew Jagielski[1]
[1]Google I

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1 Introdu

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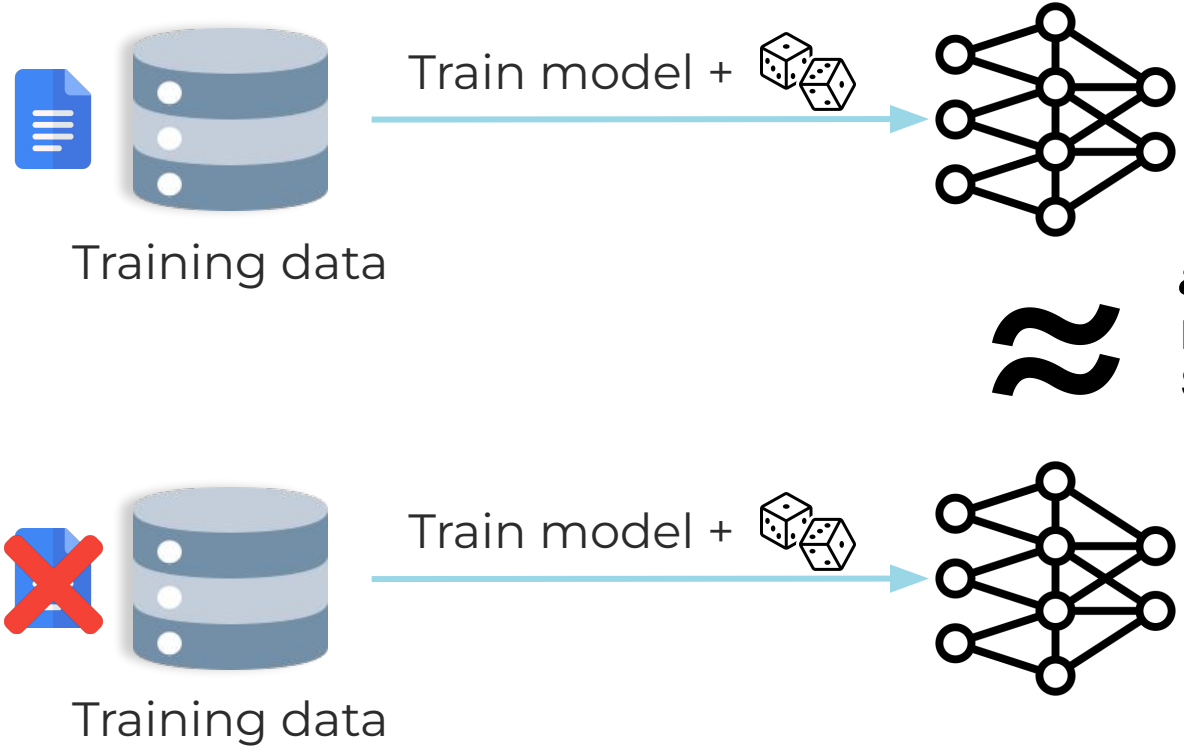
MEASURING NON-ADVERSARIAL REPRODUCTION OF TRAINING DATA IN LARGE LANGUAGE MODELS

Michael Aerni*[1] Javier Rando*[1] Edoardo Debenedetti[1]
Nicholas Carlini[2] Daphne Ippolito[2,3] Florian Tramèr[1]
[1]ETH Zurich [2]Google DeepMind [3]Carnegie Mellon University

ABSTRACT

Large language models *memorize* parts of their training data. Memorizing short snippets and facts is required to answer questions about the world and to be fluent in any language. But models have also been shown to reproduce long verbatim sequences of memorized text when prompted by a motivated adversary. In this work, we investigate an intermediate regime of memorization that we call *non-adversarial reproduction*, where we quantify the overlap between model responses and pretraining data when responding to *natural and benign prompts*. For a variety of innocuous prompt categories (e.g., writing a letter or a tutorial), we show that up to 15% of the text output by popular conversational language models overlaps with snippets from the Internet. In worst cases, we find generations where 100% of the content can be found exactly online. For the same tasks, we find that human-written text has far less overlap with Internet data. We further study whether prompting strategies can close this reproduction gap between models and humans. While appropriate prompting can reduce non-adversarial reproduction on average, we find that mitigating worst-case reproduction of training data requires stronger defenses—even for benign interactions.

Differential Privacy (DP)

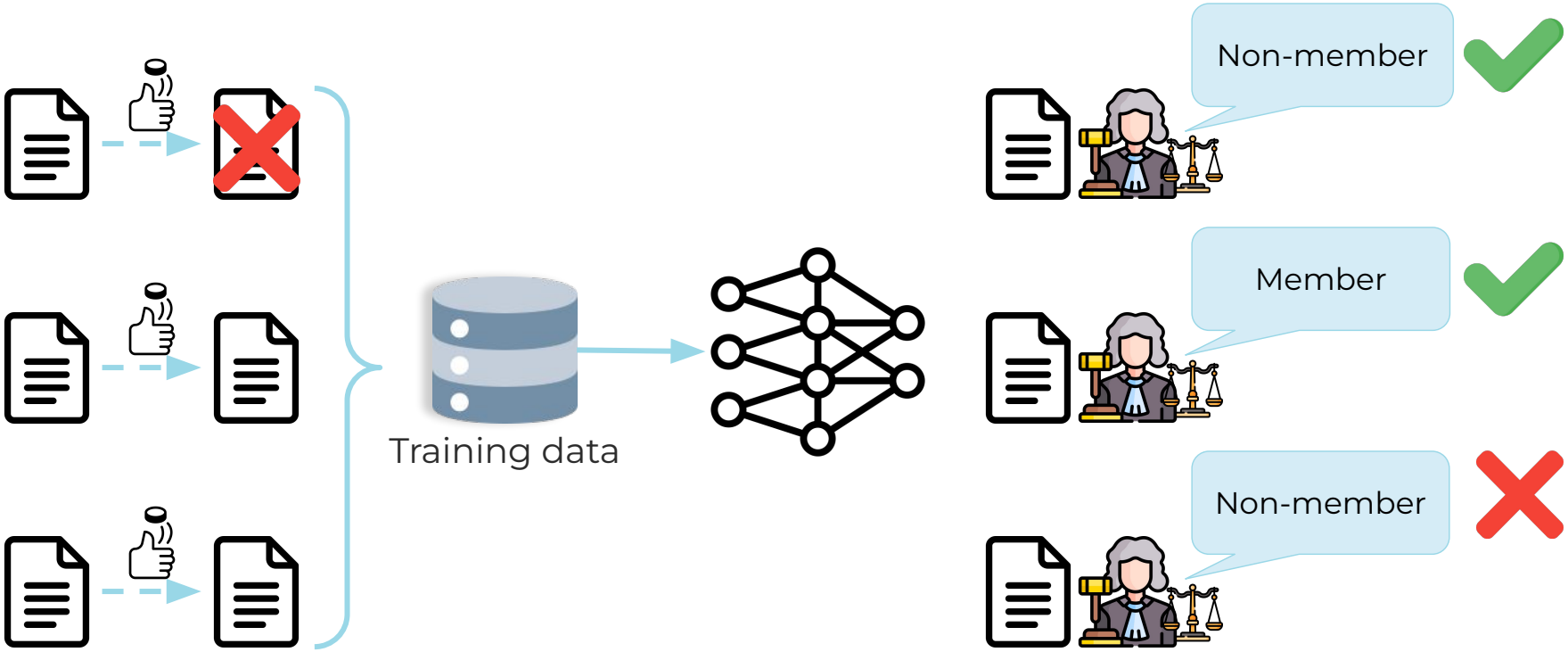


ε Privacy Budget

How close should the outputs be

Smaller ε = more privacy

Auditing Differential Privacy



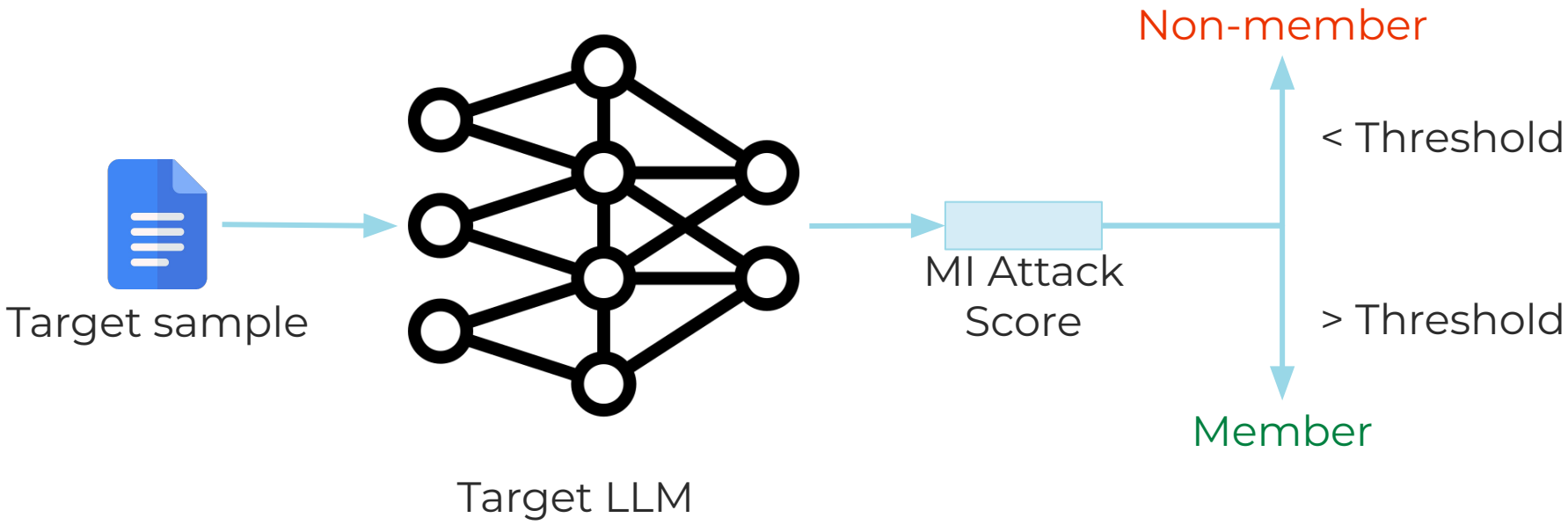
Steinke, T., Nasr, M., & Jagielski, M. Privacy auditing with one (1) training run. NeurIPS 2023

Problem: In LLMs Post-Hoc Differential Privacy Auditing Is Not Possible

Existing methods require **training at least another model**

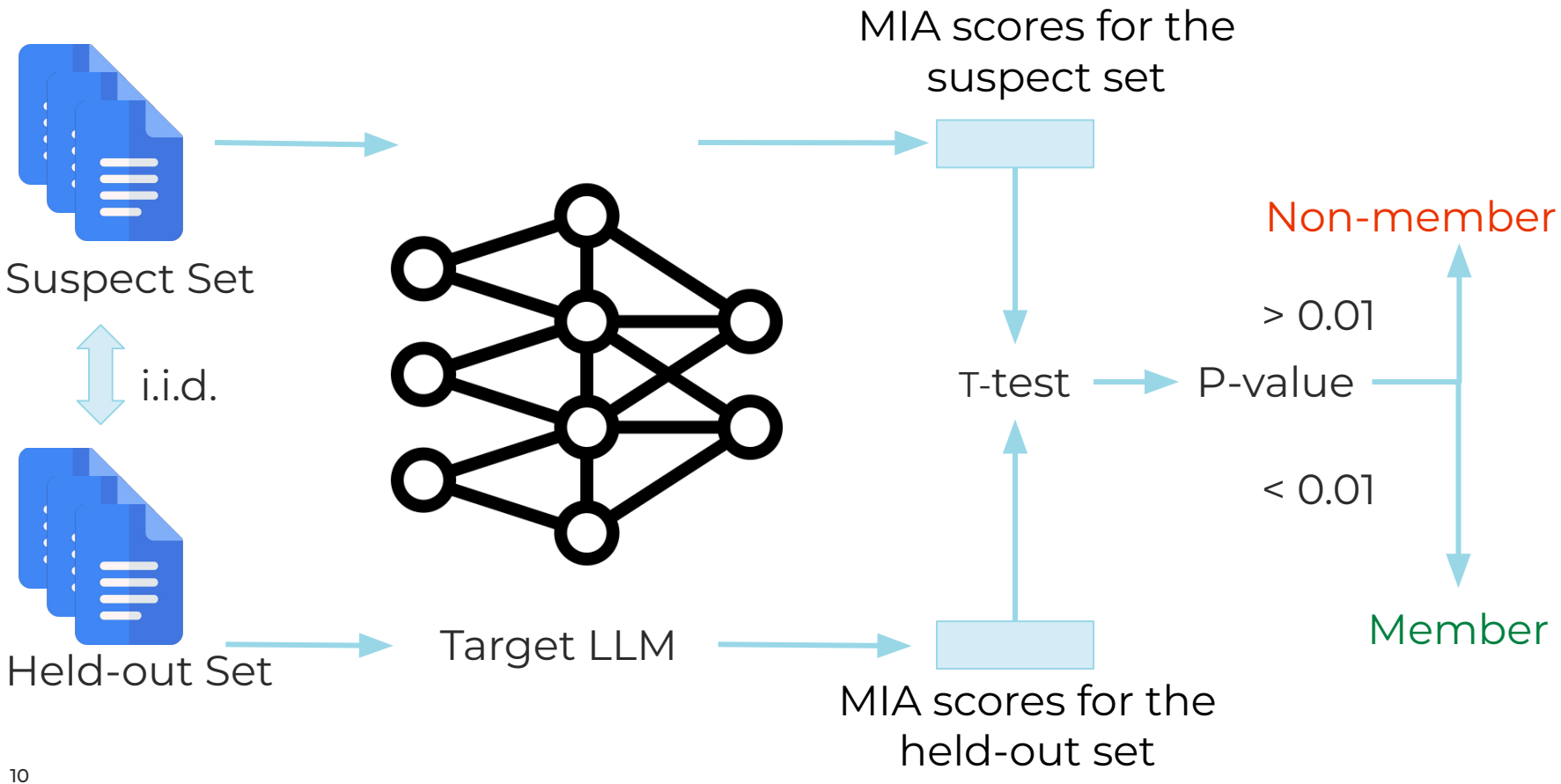
This is **impractical** for LLMs due to their **high training costs**

Membership Inference



Perplexity: Measures how confidently the model predicts a sample
MIN-K% Prob: Average likelihood over the least confident K% of tokens

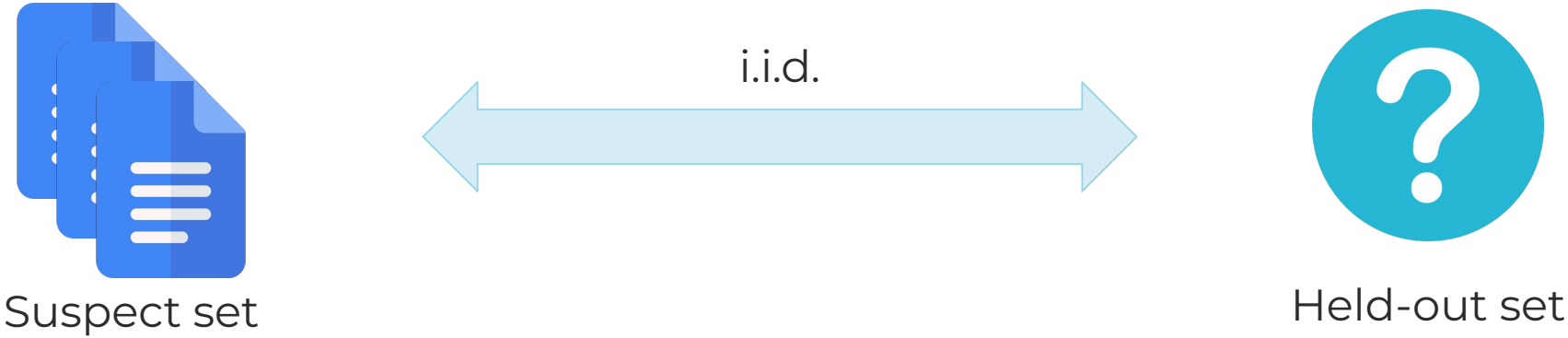
Dataset Inference



Problem: Post-Hoc Auditing

What if we don't have an i.i.d. held-out set?

- Data creators rarely keep a held-out set for audits



Unified Solution

Post-hoc differential privacy auditing without training a new model

Identifying the training data post-hoc

My Work

Solution: Look Into the Data

```
-----
b3Blk                                                    gtcn
NhAAA                                                    6jDl
...
Dmk1u                                                    K8XE
qUQ+i
-----
```

It is an SSH key

Its structure is known

Solution: Natural Identifiers (NIDs)

NIDs are structured **random strings**, generated according to some **well-defined functions**

Generated Identifiers (GIDs)

GIDs are synthetic identifiers sampled to **match the distributional properties** of NIDs but with no overlap with the training set

How To Generate GIDs


```
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAAABG5vbmUAAAAEbm9uZQAAAAAAAAABAAACFwAAAAdzc2gtcn
NhAAAAAwEAAQAAAgEA 0krC0cZ/a89rw28yO3F9RayEwUpWSTCl+KnrNJ1T/gY/PSYK6jDl
...
Dmk1uUheO2xfqEK1/DspAOnPliKJfq/fLFBcXSikDGyTjygG+6Gz54F4iiBzAl7iuPK8XE
qUQ+fZ2w8kkAAAAPbG9yZW56b0BzcHJpbnQxAQIDBA==
-----END OPENSSH PRIVATE KEY-----
```


NID A real sample from the suspect set


```
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAAABG5vbmUAAAAEbm9uZQAAAAAAAAABAAACFwAAAAdzc2gtcn
NhAAAAAwEAAQAAAgEA l8YOgpQfXbHFStrNniLHtAZWMn2JzLmL78/Rf94gU8WTbbT4HHY6
...
HZrxgtrWrOp/w8pU83l/1iHyKqZclvhkVvhFjc7WmsRbrVuXOtRBzmSJgZpdGE7awPIqpX
pA+IUG39znJ9AAAAD2xvcmVuem9Ac3ByaW50MQECAw==
-----END OPENSSH PRIVATE KEY-----
```


GID A synthetically generated sample

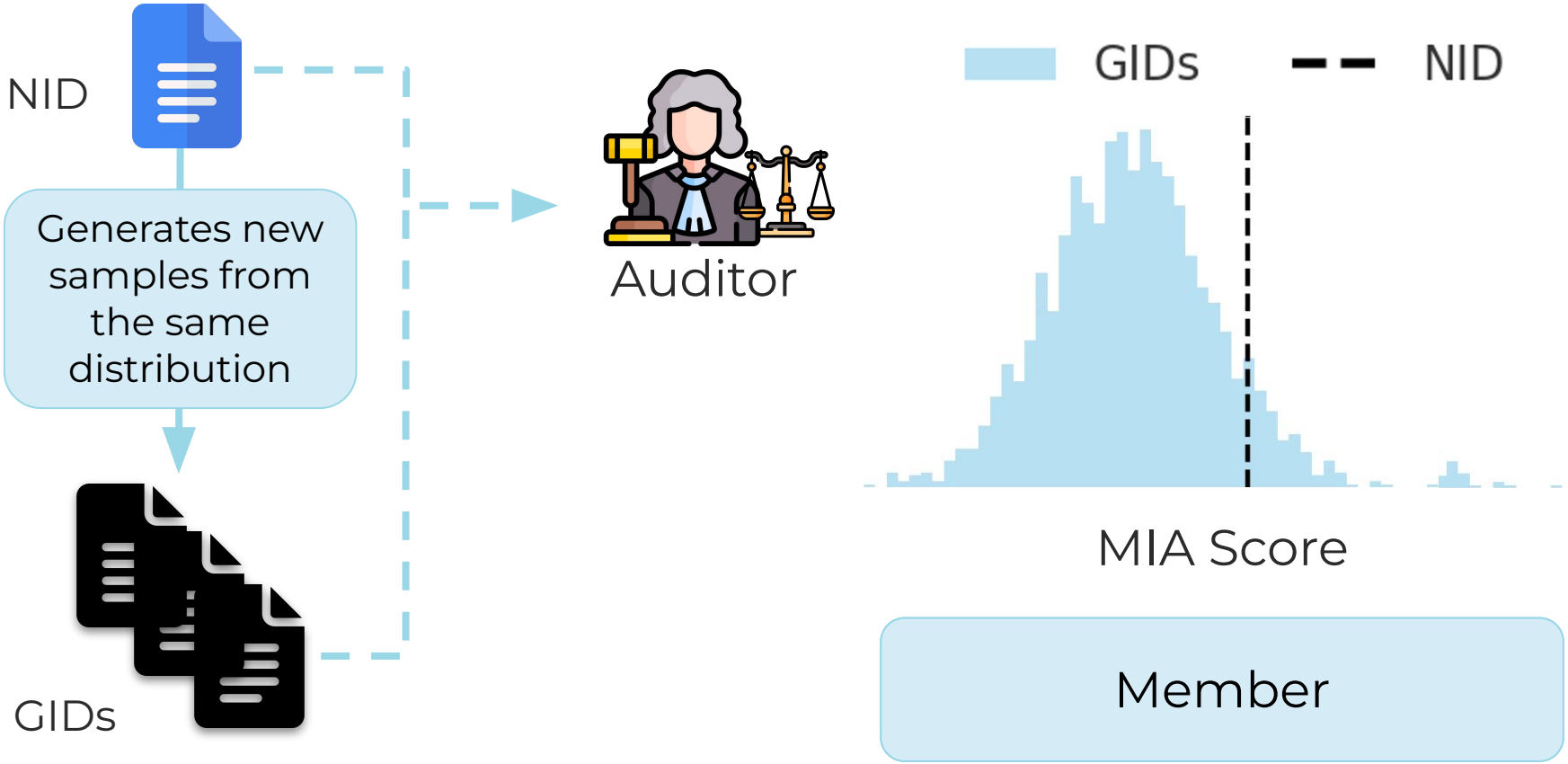
NIDs Are Everywhere

NIDs are commonly found in existing datasets such as **Pile** or **Dolma**

Types of NIDs:

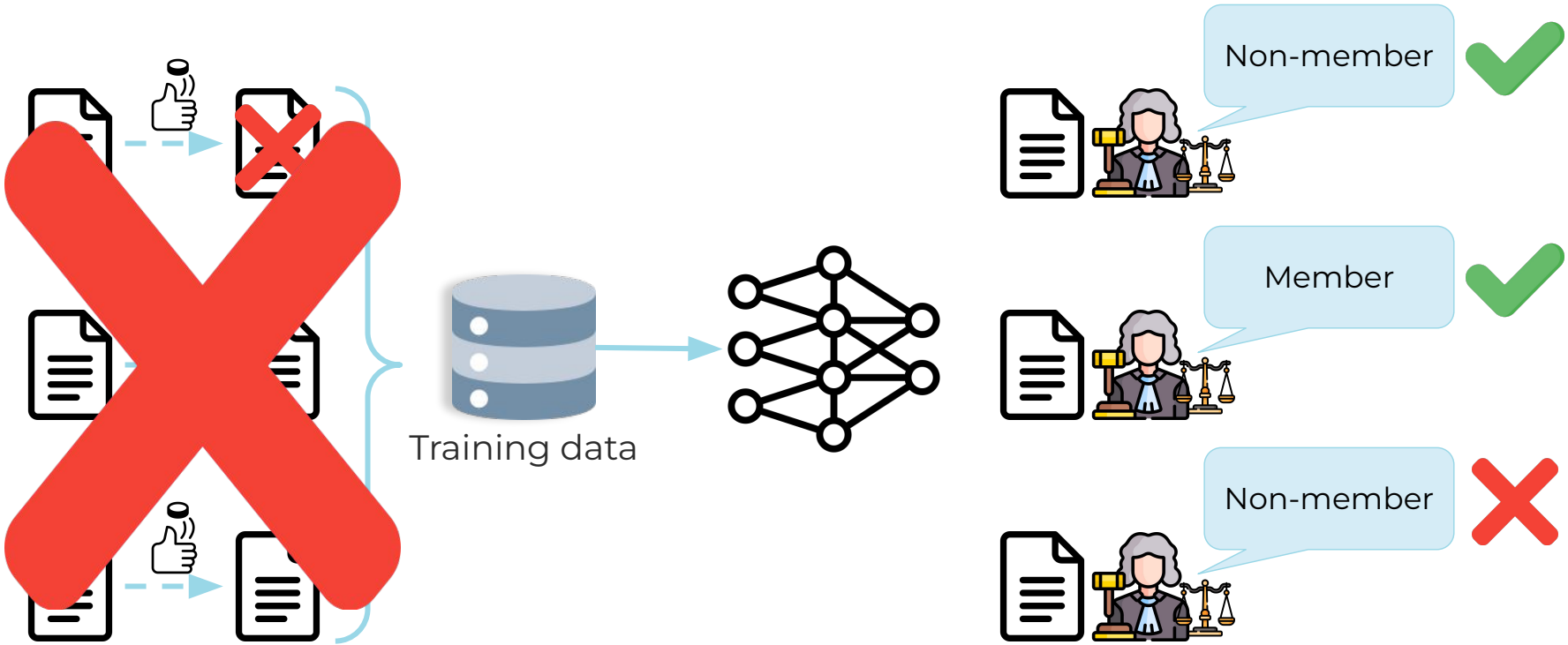
- SHA-1
- MD5
- SHA-256
- SHA-512
- Java Serialization ID
- Ethereum Address

Generates New Samples From the Same Distribution



Post-Hoc Differential Privacy Auditing

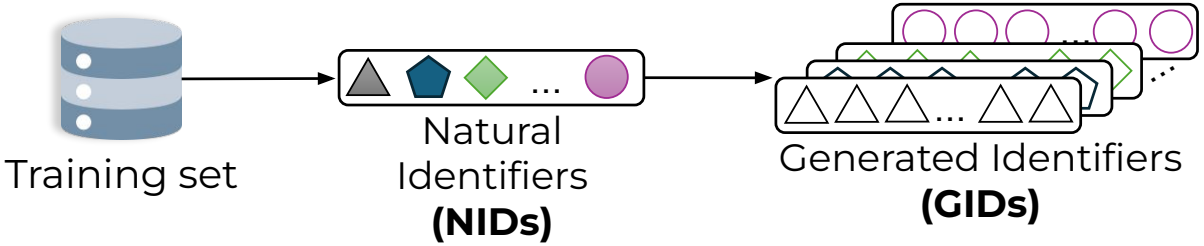
Post-Hoc Differential Privacy Auditing



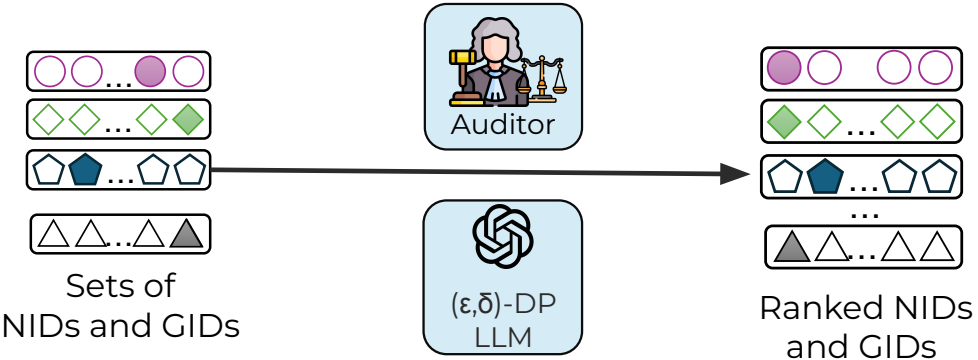
Steinke, T., Nasr, M., & Jagielski, M. Privacy auditing with one (1) training run. NeurIPS 2023

Zero Run Post-Hoc Differential Privacy Auditing

The **model owner** trains the LLM

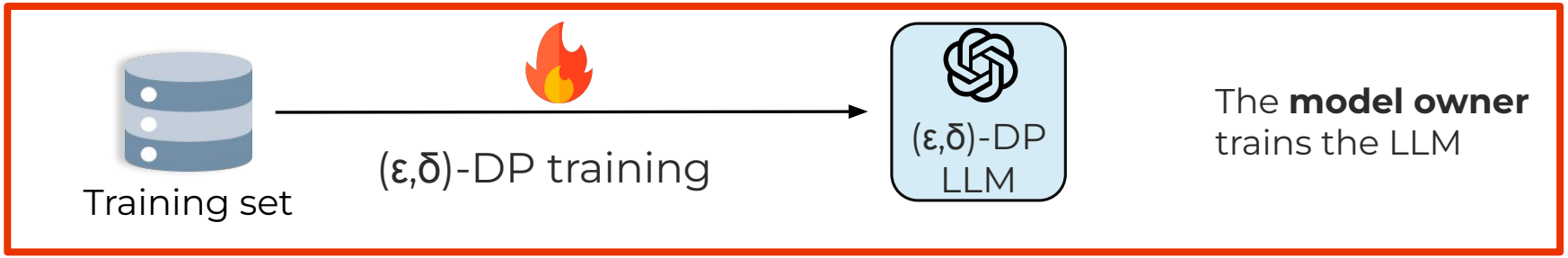


NIDs **Extraction** & GIDs **Generation**

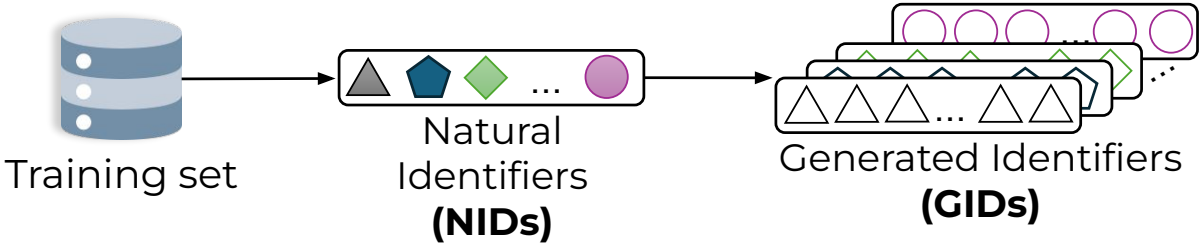


Post-hoc Differential Privacy **Auditing**

Zero Run Post-Hoc Differential Privacy Auditing



The **model owner** trains the LLM



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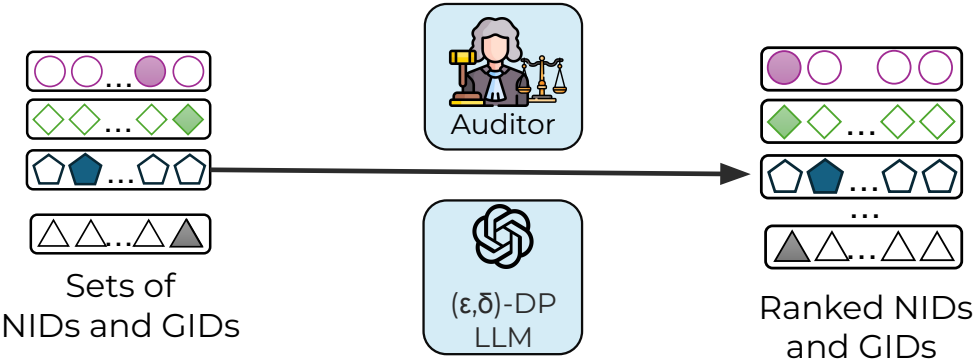
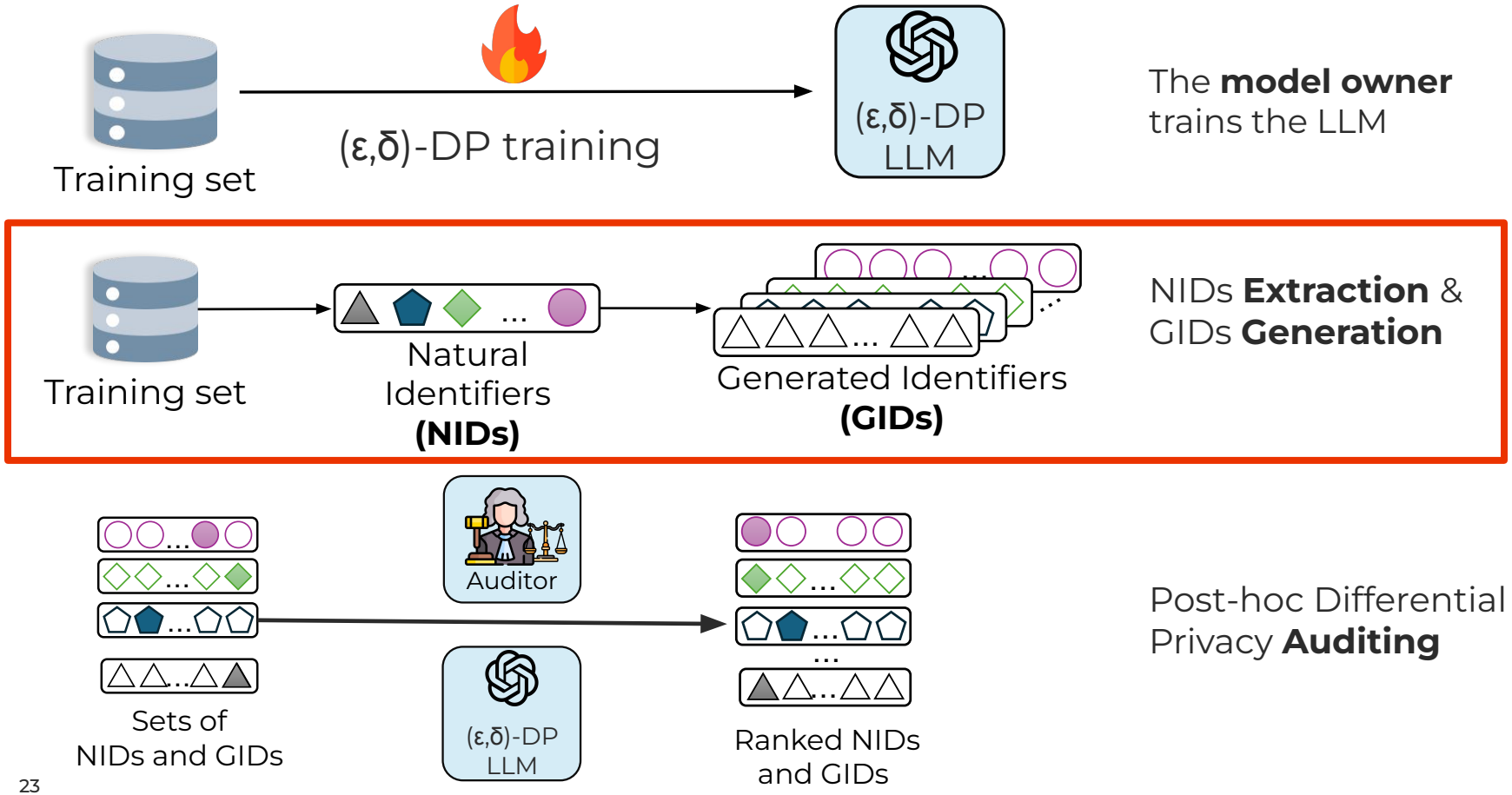


Post-hoc Differential Privacy **Auditing**

The Model Owner Trains the LLM

Zero Run Post-Hoc Differential Privacy Auditing



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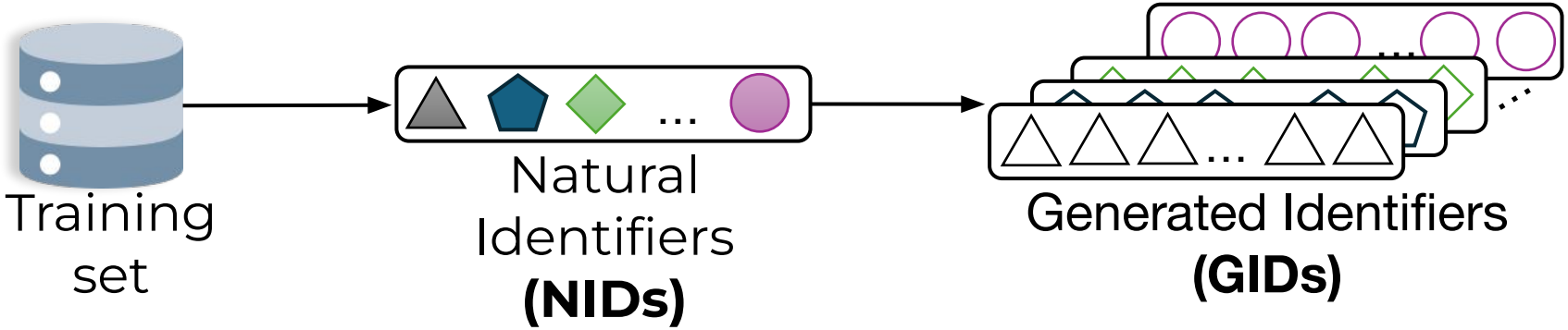

Zero Run Post-Hoc Differential Privacy Auditing

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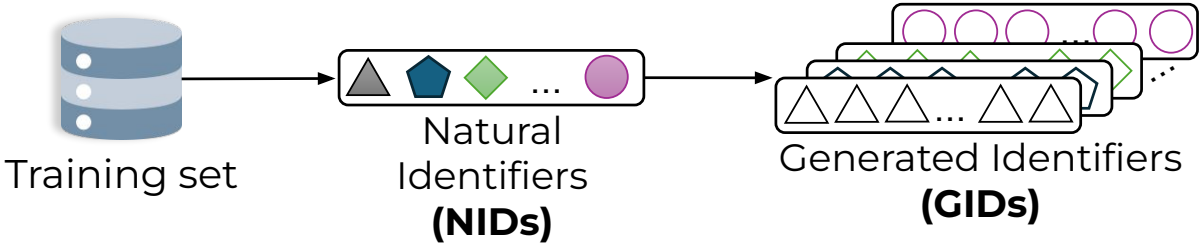

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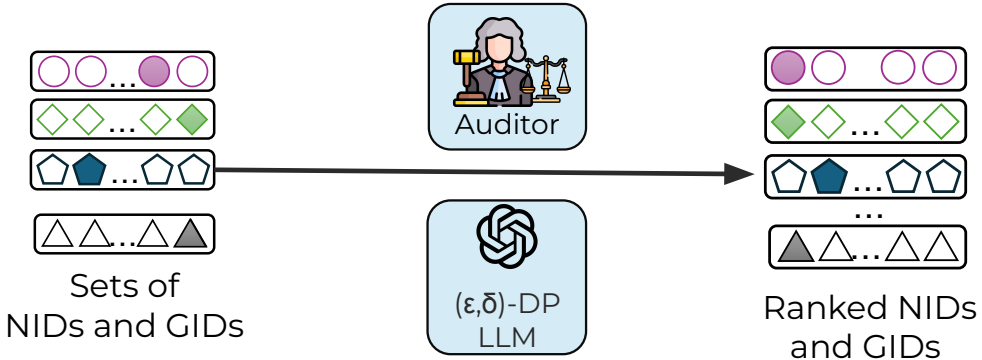
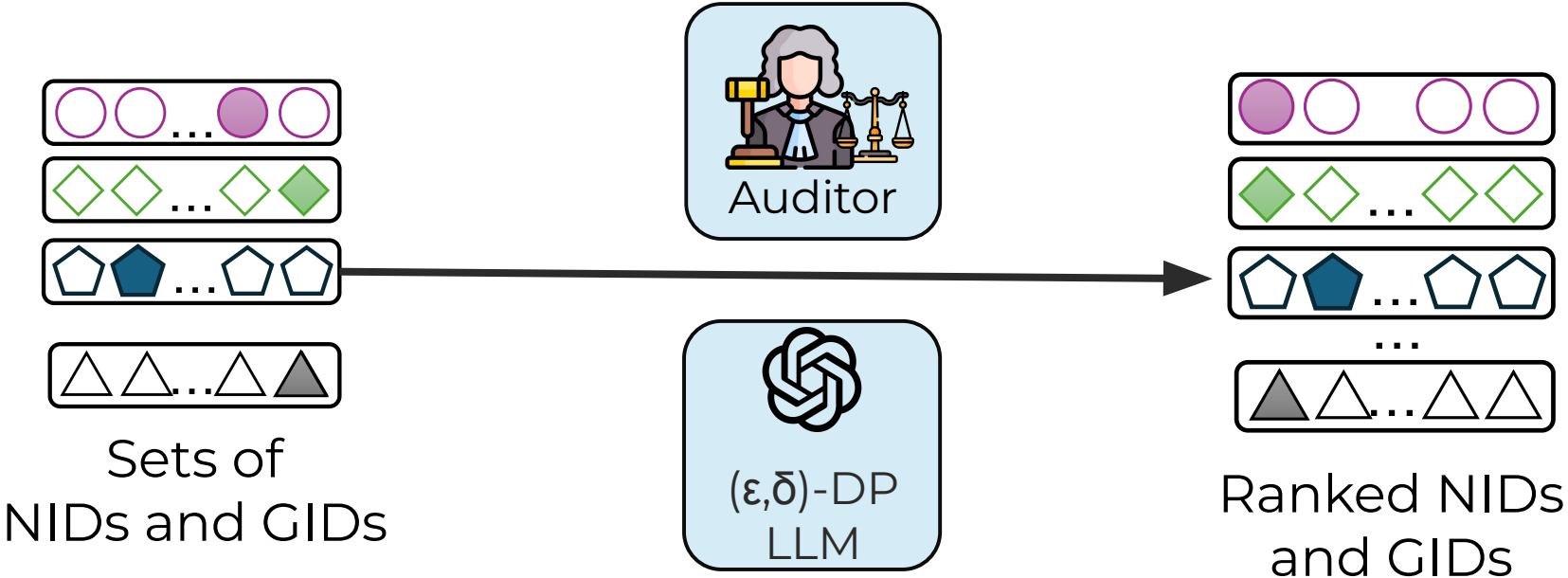


Post-hoc Differential Privacy **Auditing**

Post-hoc Differential Privacy Auditing



Theorem: Auditor Accuracy

The probability that the attacker ranks the correct **NID** in the **top-r among k options** is

$$\mathbb{P}\left[\overbrace{\overbrace{\text{⬠ ⬟ ⬠}}^{r}\text{ ⬠ ⬠ ⬠}}^{k}\right] \leq \frac{re^{\varepsilon}}{k-1+e^{\varepsilon}}$$

Previous work (Steinke et al.): k = 2, r = 1 only
Our bound: valid for any k, r

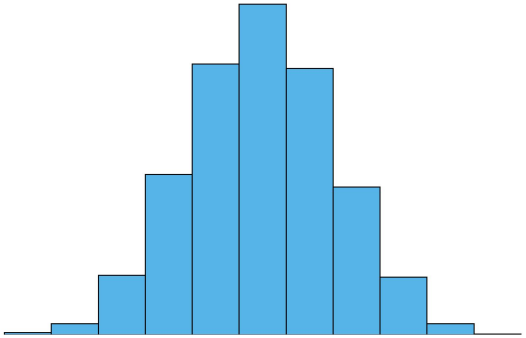
k	Number of options
r	Accuracy threshold
ε	Privacy Budget

Theorem: Auditor Accuracy

$$\mathbb{P}\left[\overbrace{\underbrace{\text{⬠ ⬟ ⬠ ⬠ ⬠ ⬠}}_{k}}^{r}\right] \leq \frac{re^{\varepsilon}}{k-1+e^{\varepsilon}}$$

$$+ \quad \cdots \quad +$$

$$\mathbb{P}\left[\overbrace{\underbrace{\text{⬠ ⬟ ⬠ ⬠ ⬠ ⬠}}_{k}}^{r}\right] \leq \frac{re^{\varepsilon}}{k-1+e^{\varepsilon}}$$

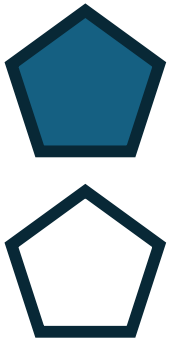


$$\leq \text{Binomial}\left(N, \frac{re^{\varepsilon}}{k-1+e^{\varepsilon}}\right)$$

N	Number of NIDs
k	Number of options
r	Accuracy threshold
ε	Privacy Budget

Why Is Having More Options Helpful?

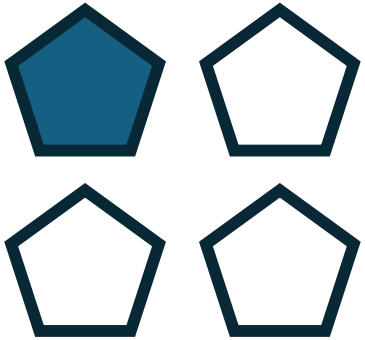
With 2 options



50% chance of guessing correctly without any knowledge of the training data

Why Is Having More Options Helpful?

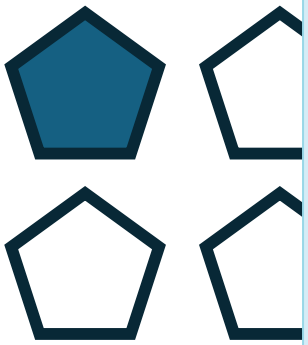
With 4 options



25% chance of guessing correctly without any knowledge of the training data

Why Is Having More Options Helpful?

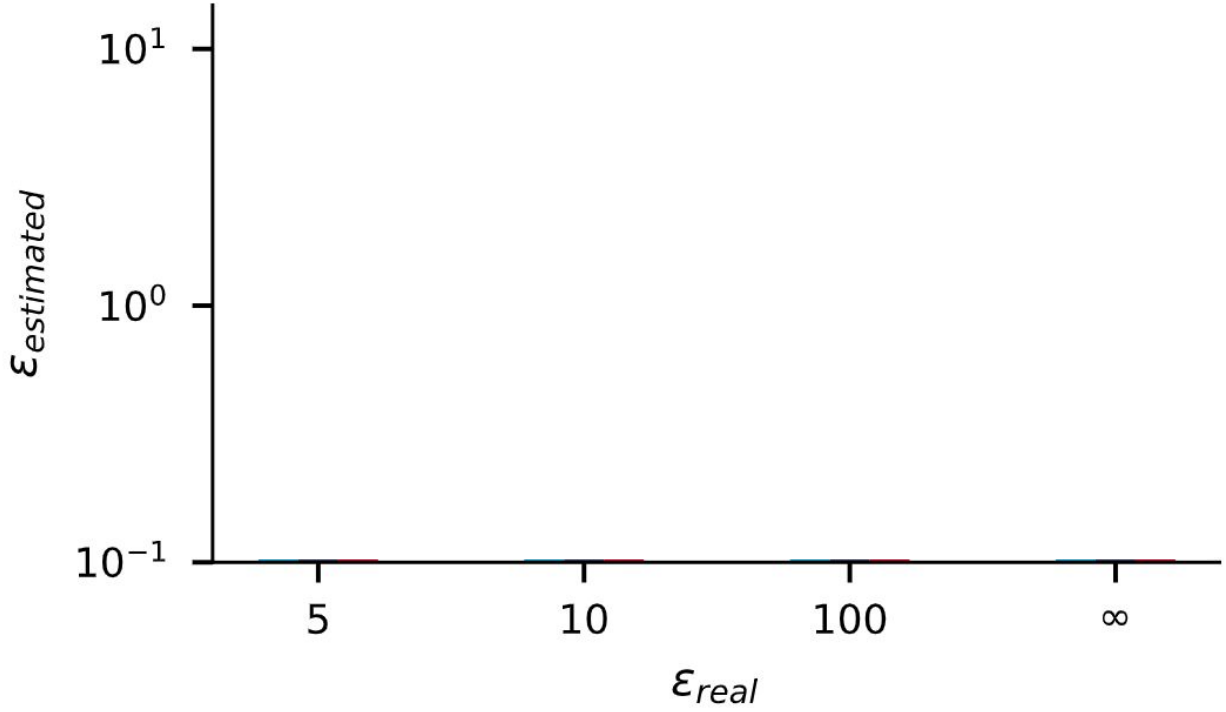
With 4 options



Correct guesses give **stronger evidence** with more options

rrectly without
ng data

Results: DP Auditing



Number of options k 2 (Steinke et al.) 8 32

Post-Hoc Dataset Inference

Dataset inference

H_0: Suspect set $\not\subset$ Training data

H_A: Suspect set $\subset$ Training data

Suspect set ⟷ i.i.d. ⟷ Held-out set

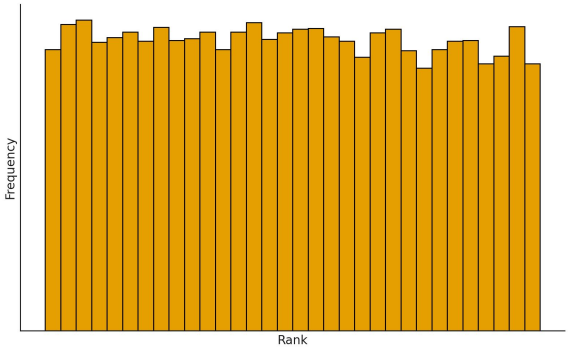
Post-Hoc Dataset Inference via Ranking

NID

H_0:

GIDs

Ranking the NID w.r.t. GIDs



Uniform distribution

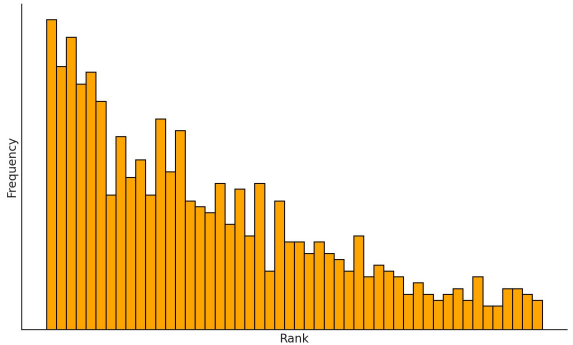
Post-Hoc Dataset Inference via Ranking

NID

H_A:

GIDs

Ranking the
NID w.r.t. GIDs

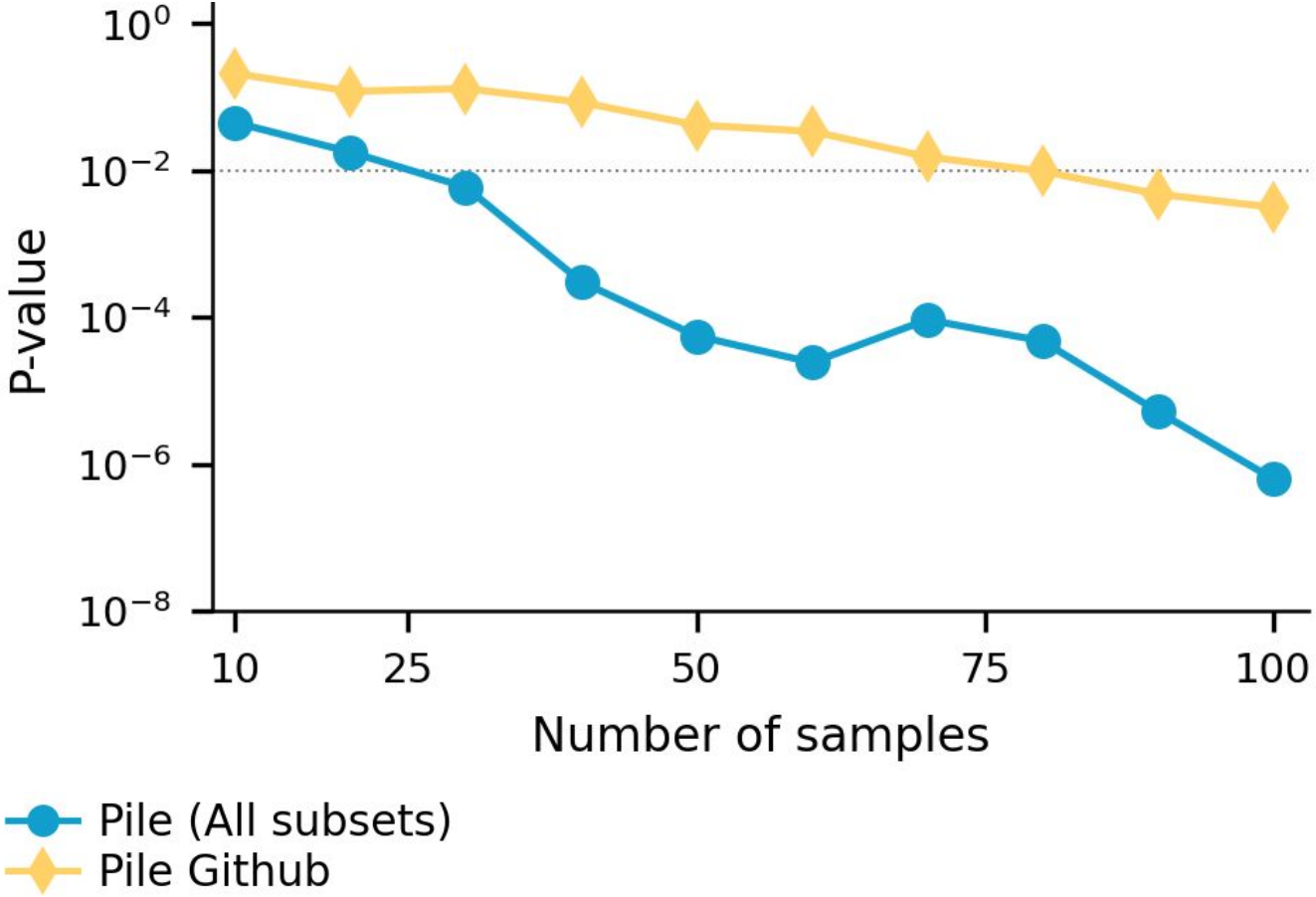


~~Uniform distribution~~

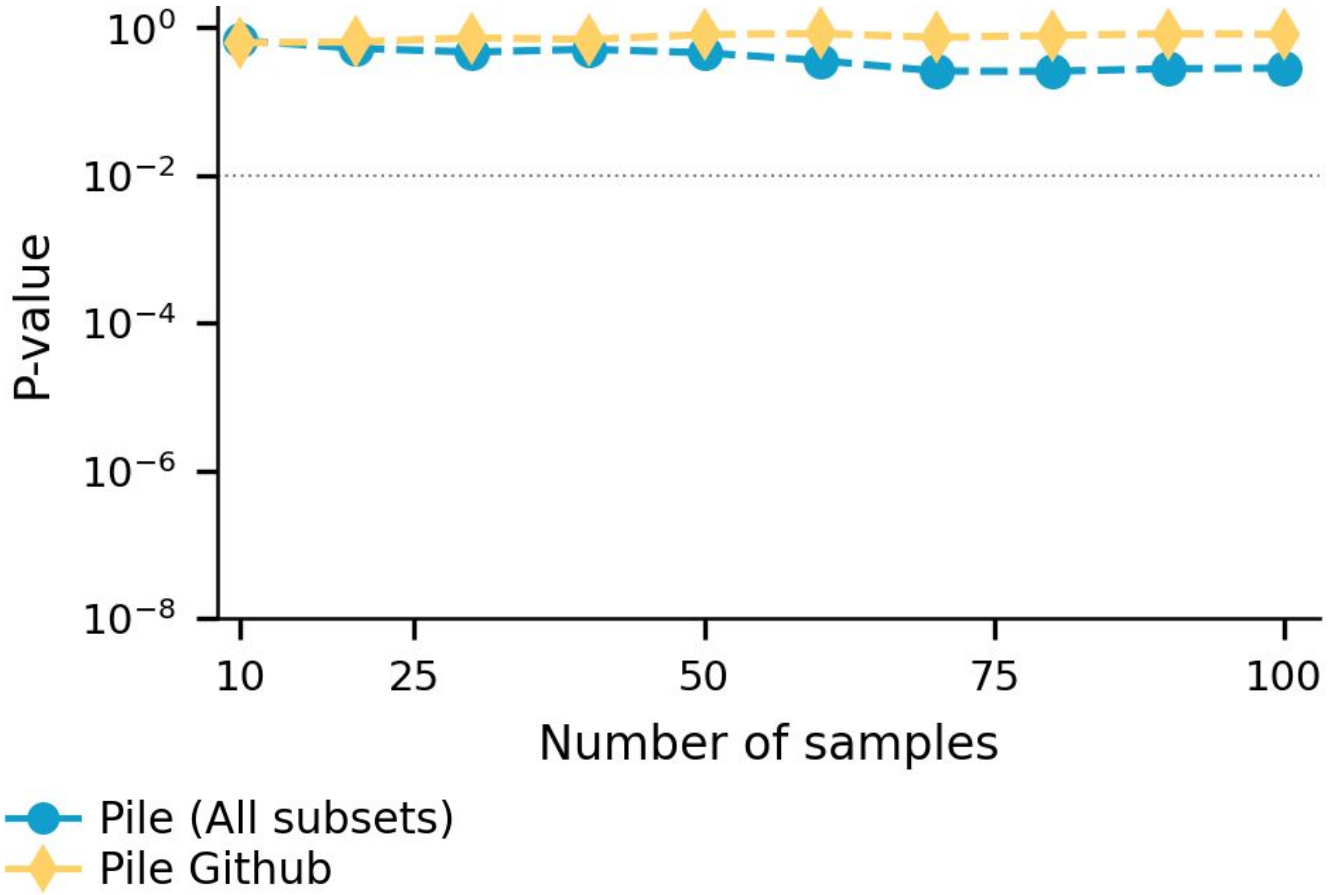
Dataset Inference: Setting

- 4 Models (Pythia 2.8B, Pythia 6.9B, Pythia 12B, OLMo 7B)
- 18 Subsets from Pile and Dolma
- 100 NIDs per subset
- 127 GIDs for each NID

Results: Training Set (Member)



Results: Test Set (Non-Member)



Unified Solution

Post-hoc differential privacy auditing without training a new model

Identifying the training data post-hoc

NIDs
Holistic post-hoc auditing

Future Vision and Ongoing Projects

Auditing Privacy in Foundation Models (FMs)

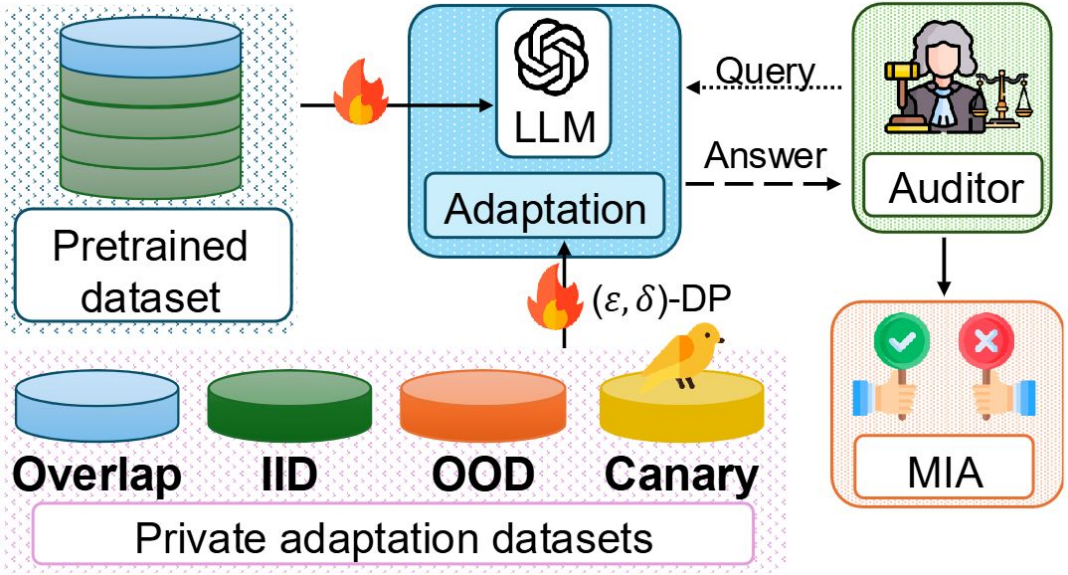
Quantify emerging forms of privacy leakage in FMs

Current Work

Future Work

Benchmarking Empirical Privacy Protection for Adaptations of Large Language Models

We analyze how data distribution and adaptation method impact privacy risks in differentially private LLM adaptations



Bartłomiej Marek , **Lorenzo Rossi****, *Vincent Hanke, Xun Wang, Michael Backes, Franziska Boenisch, Adam Dziedzic*
Under submission

Auditing Privacy in Foundation Models (FMs)

Quantify emerging forms of privacy leakage in FMs

Build scalable auditing frameworks

Current Work

Future Work

Fine-Grained and White-Box Privacy Auditing for Foundation Models

- Localizes memorization within model layers and neurons
- Moves beyond black-box membership inference
- Reveals privacy leakage concentrated in intermediate layers

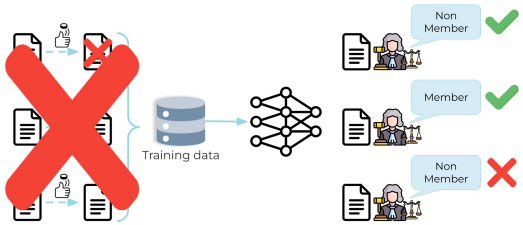
Framework **linking interpretability** and **privacy auditing**

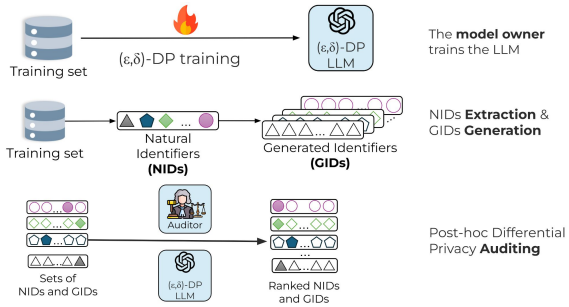
Lorenzo Rossi *, Aditya Kasliwal, Franziska Boenisch, Adam Dziedzic*
Under submission

Auditing Privacy in Foundation Models (FMs)

Current Work	Future Work
Quantify emerging forms of privacy leakage in FMs	Develop DP-based methods for multimodal FMs
Build scalable auditing frameworks	Leverage pretrain-adapt for privacy-preserving training

CISPA
HELMHOLTZ CENTER FOR INFORMATION SECURITY





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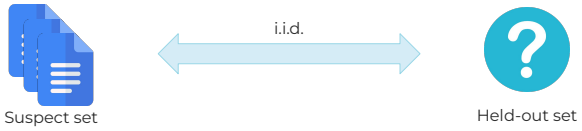


Appendix

Differential Privacy

Produce “same” outputs on any pair of datasets d and d’ that differ only by a single data point.

$$\Pr[M(d) \in S] \leq e^{\varepsilon} \Pr[M(d') \in S] + \delta$$

$\boldsymbol{M}$: Randomized mechanism, e.g., private LLM
$\boldsymbol{\varepsilon}$: How close should the outputs be. Smaller ε = more privacy
$\boldsymbol{\delta}$: Probability of closeness violation

Theorem Pure DP

Theorem 1 *Let $M : V_1 \times \cdots \times V_m \to \Sigma_1 \times \cdots \times \Sigma_m$ be an ε-DP mechanism under replacement. Let $S \in V_1 \times \cdots \times V_m$ be uniformly random, and define $T = M(S) \in \Sigma_1 \times \cdots \times \Sigma_m$. Then, for all $v \in \mathbb{R}$, all $t \in \Sigma_1 \times \cdots \times \Sigma_m$ in the support of T, all $r_1, \cdots, r_m$ with $r_i \leq |V_i|$, and $\frac{r_i e^{\varepsilon}}{|V_i|-1+e^{\varepsilon}} \leq 1$,*

$$\mathbb{P}_{\substack{S \leftarrow V_1 \times \cdots \times V_m, \\ T = M(S)}} [\sum_{i=1}^{m} \mathbb{1}[\mathrm{rank}(t_i, S_i) \leq r_i] \geq v | T = t]$$

$$\leq \mathbb{P}_{\hat{S} \leftarrow \mathrm{Bernoulli}(\frac{r_i e^{\varepsilon}}{|V_i|-1+e^{\varepsilon}})_{i=1}^{m}} [\hat{S} \geq v] := \beta(\varepsilon, v, t)$$

$\mathrm{rank}(a, b)$ *returns the 1-based position of the element b in permutation a.*

Main Theorem

Theorem 2 *Let $M : V_1 \times \cdots \times V_m \to \Sigma_1 \times \cdots \times \Sigma_m$ be an (ε, δ)-DP mechanism under replacement. Let $S \in V_1 \times \cdots \times V_m$ be uniformly random. Let $T = M(S) \in \Sigma_1 \times \cdots \times \Sigma_m$. Then, for all $v \in \mathbb{R}$, all $t \in \Sigma_1 \times \cdots \times \Sigma_m$ in the support of T, all $r_1, \ldots, r_m$ with $r_i \leq |V_i|$, and $\frac{r_i e^{\varepsilon}}{|V_i|-1+e^{\varepsilon}} \leq 1$,*

$$\mathbb{P}_{S \leftarrow V_1 \times \cdots \times V_m, T = M(S)}[\sum_{i=1}^{m} \mathbb{1}[\mathrm{rank}(t_i, S_i) \leq r_i] \geq v | T = t]$$

$$\leq \beta + \alpha\delta \sum_{i=1}^{m} |V_i|$$

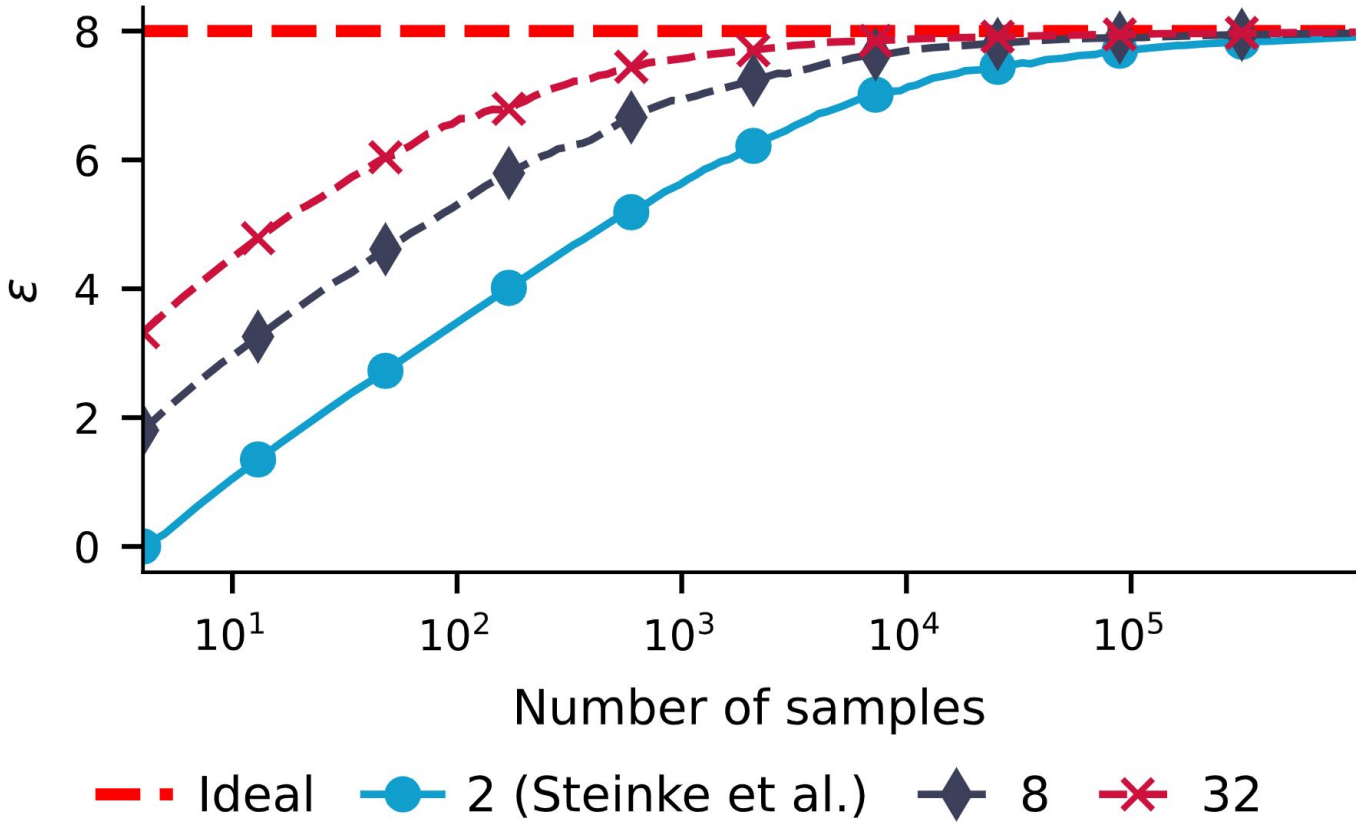
where

$$\beta = \mathbb{P}_{\hat{S}}[\hat{S} \geq v],$$

$$\alpha = \max\left(\frac{1}{i}\mathbb{P}_{\hat{S}}[\hat{S} \geq v - i] : i \in \{1, \ldots, m\}\right),$$

$$\hat{S} \leftarrow \mathrm{Bernoulli}\left(\frac{r_i e^{\varepsilon}}{|V_i| - 1 + e^{\varepsilon}}\right)_{i=1}^{m}.$$

Randomized Response



NIDs in Different Datasets

Dataset	Total Number	SHA-1	MD5	SHA-256	Java Serialization	SHA-512	Ethereum Address
dolma RefinedWeb	16989	8098	6192	2130	42	110	417
pile train github	13182	5389	1938	4158	819	701	177
pile train stackexchange	9862	4850	3235	1200	348	121	108
pile train pile cc	3422	1078	2008	274	1	8	53
dolma algebraic stack train	2384	1264	464	612	1	28	15
pile train hackernews	2268	1340	821	93	0	7	7
dolma openwebmath train	2207	1212	727	221	1	20	26
pile train ubuntuirc	1056	618	340	88	0	9	1
dolma c4	791	408	301	63	0	4	15
dolma PeS2o	435	235	174	11	0	1	14
dolma MegaWika	383	115	200	62	0	2	4
dolma ArXiv	332	239	58	21	0	2	12
Pile test (all subsets)	293	130	69	62	13	14	5
pile train pubmedcentral	225	66	152	7	0	0	0
pile train ArXiv	207	75	122	7	0	0	3
pile test github	197	80	36	52	13	12	4
pile train wikipediaen	85	15	66	3	0	1	0
pile test stackexchange	58	34	16	6	0	2	0
openwebmath test	46	19	20	6	0	1	0
algebraic stack test	39	28	4	7	0	0	0
dolma wiki	38	11	22	3	0	2	0
pile test pile cc	18	6	8	3	0	0	1
pile train philpapers	16	1	15	0	0	0	0
pile train freelaw	15	1	14	0	0	0	0
pile test hackernews	13	7	6	0	0	0	0
dolma tulu flan	10	0	9	1	0	0	0
pile test ubuntuirc	5	3	2	0	0	0	0
pile train enronemails	4	0	4	0	0	0	0
pile test wikipediaen	2	0	1	1	0	0	0
dolma books	2	0	2	0	0	0	0
pile train gutenbergpg 19	1	0	1	0	0	0	0
pile train pubmedabstracts	1	0	1	0	0	0	0

Concentration Bound

$$p = \frac{re^{\varepsilon}}{c - 1 + e^{\varepsilon}}$$

$\hat{S} \sim \text{Binomial}(m, p)$ and $\hat{S}/m$ represents the empirical accuracy

Bernstein inequality to this binomial distribution yields the following tail bound:

$$\mathbb{P}\left[\frac{\hat{S}}{m} \geq p + \frac{1}{3m}\log\left(\frac{1}{\beta}\right) + \sqrt{\frac{1}{9m}\log^2\left(\frac{1}{\beta}\right) + \frac{2p(1-p)}{m}\log\left(\frac{1}{\beta}\right)}\right] \leq \beta$$

$$\sqrt{\frac{2p(1-p)}{m}\log\left(\frac{1}{\beta}\right)} \qquad \Theta\left(\sqrt{\frac{1}{mc}}\right)$$

NID

For a future birthday party – fairy party favors. But I want to figure out a different fairy, not Disney...**34d42a69a258fa51222a2e94b4563007**.jpg 300×300 pixels
A quick, easy project for the kids: playful, pom-pom covered trees.
Carrot & Apple Cinnamon Streusel Muffins | a cup of mascarpone
Strawberry Banana Muffins recipe

Identifying the Training Data Post-Hoc

Membership Inference Attack (MIA)

Target sample

i.i.d.

Held-out set

Dataset Inference (DI)

Suspect set

i.i.d.

Held-out set